



# GLAXIT

INTERNSHIP PROGRAM

*Advanced Cyber Security*

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## Task 3 Report

*Basic Security Monitoring — Linux System (Kali Linux VM)*

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## Basic Security Monitoring — Linux System (Kali Linux VM)

Routine defensive monitoring was performed on a Kali Linux VM by reviewing login history, running processes, listening network ports, and system/authentication logs. Two suspicious indicators were identified: a burst of failed local sudo authentication attempts, and a login session from an unrecognized user account. Running processes and listening ports were reviewed and found to be consistent with expected system behavior. Screenshots of each command's output are included as evidence below.

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date &amp; Time Observed</b> | Aug 15, 2026, 05:44:19 – 05:46:24 (local)                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Command / Log Checked</b>    | journalctl -p err -b                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Suspicious Activity</b>      | Five separate sudo authentication failures within a ~2 minute window, each recording 3 incorrect password attempts (15 failed attempts total), while trying to run privileged commands (ss -tulnp, journalctl -u ssh -u sshd).                                                                                                                                                                                                                                                  |
| <b>Supporting Evidence</b>      | See Figure 1 below — journalctl error-level output.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Possible Security Risk</b>   | Repeated, rapid sudo failures are a classic indicator of a local privilege-escalation / brute-force attempt against an account. Even if this turns out to be the legitimate user mistyping a password, the same log pattern is what an attacker or malicious script guessing sudo credentials would produce, and it warrants confirmation rather than being dismissed.                                                                                                          |
| <b>Recommended Action</b>       | <p>Confirm with the account owner whether these attempts were legitimate.</p> <p>Check fail2ban configuration to ensure repeated auth failures trigger a lockout (fail2ban-server is already running on this host — verify it covers PAM/sudo, not just SSH).</p> <p>Review /var/log/auth.log around this timestamp for the originating session and TTY.</p> <p>If not attributable to the legitimate user, change the account password and audit sudoers group membership.</p> |

[illegible]

Figure 1: `journalctl -p err -b` — repeated "3 incorrect password attempts" entries between 05:44:19 and 05:46:24

## Finding 2: Unrecognized User Account Login

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date &amp; Time Observed</b> | Sat Aug 8, 2026, 13:47 – 13:51 (login session)                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Command / Log Checked</b>    | sudo last   head -20                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Suspicious Activity</b>      | A login session by user account "HasnainA" appears in the login history via console tty7, alongside only the expected accounts (kali, lightdm, root) seen in every other entry. This account is not part of the system's normal/expected user list.                                                                                                                                                                                                                                        |
| <b>Supporting Evidence</b>      | See Figure 2 below — last command output.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Possible Security Risk</b>   | An unrecognized local account with console login activity may indicate an unauthorized user account was created (or an existing test/shared account was used) and someone other than the primary user accessed the system directly at the console.                                                                                                                                                                                                                                         |
| <b>Recommended Action</b>       | Verify with the system owner whether "HasnainA" is a known/authorized account (e.g., a classmate sharing the VM, a lab account).<br>Run <code>cat /etc/passwd</code> and <code>id HasnainA</code> to check the account's privileges and creation date.<br>If unauthorized, disable or delete the account ( <code>sudo userdel -r HasnainA</code> ) and rotate credentials for all accounts.<br>Enable auditd or ongoing last/lastlog monitoring to catch future unexpected console logins. |

```
(kali@kali)-[~]
$ sudo last | head -20
kali      tty7      :0                Sat Aug 15 05:32 - still logged in
lightdm   tty7      :0                Sat Aug 15 05:32 - 05:32 (00:00)
kali      tty7      :0                Sat Aug 15 05:18 - still logged in
lightdm   tty7      :0                Sat Aug 15 05:18 - 05:18 (00:00)
kali      tty7      :0                Tue Aug 11 06:51 - still logged in
lightdm   tty7      :0                Tue Aug 11 06:51 - 06:51 (00:00)
kali      tty7      :0                Tue Aug 11 05:23 - still logged in
lightdm   tty7      :0                Tue Aug 11 05:23 - 05:23 (00:00)
kali      tty7      :0                Tue Aug 11 05:13 - still logged in
lightdm   tty7      :0                Tue Aug 11 05:13 - 05:13 (00:00)
kali      tty7      :0                Sat Aug 8 13:51 - still logged in
lightdm   tty7      :0                Sat Aug 8 13:47 - 13:51 (00:04)
HasnainA  tty7      :0                Sat Aug 8 06:41 - still logged in
lightdm   tty7      :0                Sat Aug 8 06:40 - 06:41 (00:00)
root      pts/0     :0                Sat Aug 8 06:40 - 06:40 (00:00)
kali      tty7      :0                Sat Aug 8 06:38 - 06:40 (00:02)
lightdm   tty7      :0                Sat Aug 8 06:37 - 06:38 (00:00)
root      pts/0     :0                Sat Aug 8 06:37 - 06:37 (00:00)
kali      tty7      :0                Sat Aug 8 06:17 - 06:37 (00:20)
lightdm   tty7      :0                Sat Aug 8 06:17 - 06:17 (00:00)
```

Figure 2: `sudo last | head -20` — "HasnainA" login session on Aug 8, 13:47–13:51

## Additional Checks Performed (No Anomalies Found)

Running processes, listening network ports, and the authentication log were also reviewed. Nothing unexpected was found in these areas — included below for completeness.

### Running Processes

ps aux --sort=-%cpu and top -b -n1 showed only expected processes: Xorg, VBoxClient, xfce4-panel plugins, systemd/kworkers, containerd, dockerd, and fail2ban-server. No unfamiliar binaries, no processes running from unusual paths (e.g. /tmp), and no abnormal CPU/memory consumption.

```
(kali@kali)-[~]
└─$ ps aux --sort=-%cpu | head -20
USER        PID %CPU %MEM    VSZ   PTY  STAT START   TIME COMMAND
root      11878 50.0  0.1  9604 4676 pts/0  R+   05:50   0:00 ps aux --sort=-%cpu
root       713  6.2  0.4 511816 374808 ttyp  Ssl   05:32   1:00 /usr/lib/serg/lorg -d -seat seat0 -auth /var/run/lightdm/root/:0 -nolisten tcp vt7 -novtswitch
root       697  2.9  1.1 1492120 47396 ?    Ssl   05:32   0:33 /usr/bin/containerd
kali      8151 1.0  0.1 186408 6604 pts/0  Ss   05:44   0:07 /usr/bin/fish
kali      8134 1.0  1.8 667608 74952 ?    Sl    05:44   0:07 /usr/bin/terminal
kali     1528 1.5  1.2 308912 49984 ?    Sl    05:32   0:17 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /usr/lib/x86_64-linux-gnu/xfce4/panel/plugins/libcpugraph.so 13 25165836 cpugraph CPU Graph Graphical representation o
f the CPU load
kali     1339 0.9  0.0 212768 3172 ?    Sl    05:32   0:10 /usr/bin/VBoxClient --draganddrop
kali     2489 0.9  2.1 300828 362480 ?    Sl    05:32   0:18 xfwm
kali     1523 0.8  0.7 277048 38648 ?    Sl    05:32   0:08 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /usr/lib/x86_64-linux-gnu/xfce4/panel/plugins/libgmon.so 15 25165818 gmon Generic Monitor Show output of a command
,
root     1398 0.2  0.2 318944 11024 ?    Ssl   05:32   0:06 /usr/bin/python3 /usr/bin/fail2ban-server -xf start
root     1518 0.2  0.2 318944 11024 ?    Ssl   05:32   0:02 /usr/libexec/upowerd
root     1518 0.2  0.0 318944 11024 ?    Ssl   05:32   0:02 /usr/bin/ModemManager
kali     1498 0.2  1.5 1193588 62368 ?    Sl    05:32   0:02 xfdesktop
root       1  0.2  0.4 26816 16552 ?    Ss   05:32   0:02 /sbin/init splash
root       735  0.2  1.9 154968 77492 ?    Ssl   05:32   0:02 /usr/sbin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
kali     1489 0.2  1.1 917928 46120 ?    Sl    05:32   0:02 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /usr/lib/x86_64-linux-gnu/xfce4/panel/plugins/libwhiskermenu.so 1 25165811 whiskermenu Whisker Menu Show a menu to eas
ily access installed applications
root     4983 0.2  0.0 0 0 ?        I    05:37   0:01 [worker/0:2-events]
messages  504  0.2  0.1 186408 7200 ?    Ss   05:32   0:02 /usr/bin/dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
kali     1457 0.1  1.3 985888 53284 ?    Sl    05:32   0:02 xfce-panel
```

Figure 3: ps aux --sort=-%cpu output

```
(kali@kali)-[~]
└─$ top -b -n1 | head -20
top - 05:51:22 up 19 min, 1 user, load average: 0.42, 0.66, 0.57
Tasks: 184 total, 1 running, 183 sleep, 0 d-sleep, 0 stopped, 0 zombie
%Cpu(s): 5.0 us, 7.5 sy, 0.0 ni, 87.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3917.3 total, 1036.8 free, 1052.2 used, 2124.7 buff/cache
MiB Swap: 953.7 total, 953.7 free, 0.0 used. 2865.1 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
    713 root       20   0 511816 178060 87296 S   9.4   4.4   1:10.56 Xorg
   1339 kali      20   0 212768   3372  2976 S   4.7   0.1   0:10.69 VBoxCli+
   1520 kali      20   0 300912 49984 22292 S   4.7   1.2   0:17.56 wrapper+
    1 root       20   0 26816 16552 12256 S   0.0   0.4   0:02.64 systemd
    2 root       20   0 0 0 0 S   0.0   0.0   0:00.08 kthreadd
    3 root       20   0 0 0 0 S   0.0   0.0   0:00.00 pool_wo+
    4 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker+
    5 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker+
    6 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker+
    7 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker+
    8 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker+
   11 root       20   0 0 0 0 I   0.0   0.0   0:02.07 kworker+
   12 root       20   0 0 0 0 I   0.0   0.0   0:00.00 kworker+
```

Figure 4: top -b -n1 output

### Listening Network Ports

sudo ss -tulnp showed a single listening socket: 127.0.0.1:41991 (containerd). This is bound to loopback only and is not reachable from the network, so it does not represent an exposed/unusual open port.

```
(kali@kali)-[~]
└─$ sudo ss -tulnp
Netid  State     Recv-Q    Send-Q           Local Address:Port           Peer Address:Port           Process
tcp    LISTEN          0      4096             127.0.0.1:41991              0.0.0.0:*                  users(("containerd",pid=697,fd=10))
```

Figure 5: sudo ss -tulnp output

### Authentication Log (Failed Passwords)

grep "Failed password" /var/log/auth.log returned no genuine failed-SSH-login entries — only the sudo audit record of running the grep command itself. No evidence of remote SSH brute-force activity was found in this log.

```
(kali@kali)-[~]
└─$ sudo grep "Failed password" /var/log/auth.log | tail -n 20
2026-08-11T05:27:33.225096-04:00 kali sudo:      kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/grep 'Failed password' /var/log/auth.log
2026-08-15T05:49:03.165238-04:00 kali sudo:      kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/grep 'Failed password' /var/log/auth.log
2026-08-15T05:49:25.042803-04:00 kali sudo:      kali : TTY=pts/0 ; PWD=/home/kali ; USER=root ; COMMAND=/usr/bin/grep 'Failed password' /var/log/auth.log
```

Figure 6: grep "Failed password" /var/log/auth.log output

## Overall Conclusion

The most significant indicator from this monitoring session is the cluster of sudo authentication failures, which — combined with an unfamiliar account appearing in login history — should be verified with the system owner. Neither finding conclusively proves compromise, but both meet the threshold for follow-up under a defensive monitoring process rather than being dismissed as noise. Process, network, and authentication-log checks did not surface additional indicators at this time.